Apportionment Paradoxes: The Balinski-Young Impossibility Theorem and Divisor Method Immunity

Giovanni Della-Libera

May 8, 2026

Abstract

The Alabama Paradox — in which increasing the House size causes a state to *lose* a seat — was documented in the 1880 Census and is the most politically embarrassing property of the Hamilton (largest remainders) apportionment method. Two related paradoxes complete the triad of classical apportionment anomalies: the Population Paradox (a faster-growing state loses a seat to a slower-growing state) and the New-States Paradox (adding a state with proportional seats causes existing states to exchange seats). All three paradoxes afflict Hamilton but not the divisor methods (Huntington-Hill, Webster, Adams, Jefferson).

This paper proves, with formal mathematics and numerical examples, (1) that Hamilton is susceptible to all three paradoxes, (2) that all four divisor methods are immune to all three paradoxes, and (3) that no apportionment method can be simultaneously quota-compliant and paradox-immune — the Balinski-Young impossibility theorem (1982).

For Hamilton susceptibility we give numerical examples: the 1880 Alabama Paradox (adding a seat reduced Alabama from 8 to 7), a 1900 Population Paradox example from U.S. census history, and the 1907 Oklahoma New-States Paradox. For divisor method immunity we give a formal proof using the monotone priority-queue structure: in a priority-queue divisor method, adding a seat or increasing a state's population can only add seats — no existing seat allocation is reconsidered. The Balinski-Young theorem is proved by showing that satisfying the quota rule for all possible population vectors forces susceptibility to the Alabama paradox: the fractional-remainder ranking that ensures quota compliance is necessarily non-monotone in House size. The paper concludes with the historical and policy consequences of these mathematical results for Congress's 1941 adoption of Huntington-Hill.

Keywords

Alabama paradox, population paradox, new-states paradox, Hamilton method, Balinski-Young impossibility theorem, divisor methods, apportionment, quota rule, paradox immunity

1 Introduction: Three Paradoxes in U.S. Apportionment History

The mathematical study of apportionment began in earnest with a paradox. In 1880, the chief clerk of the Census Bureau, C. W. Seaton, computed apportionments for House sizes ranging from 275 to 350. He discovered that when the House was sized at 299, Alabama received 8 seats; when sized at 300, Alabama received only 7. The addition of a seat to the House had caused a state to *lose* a seat.

This result, which came to be called the *Alabama Paradox*, was not a computational error. It was a mathematical property of the Hamilton method — the method then in use for U.S. congressional apportionment. Seaton wrote to members of Congress to report the finding, and it triggered a political crisis: if the House size could not be set without risking arbitrary redistributions of seats, the entire apparatus of periodic reapportionment was in question.

The Alabama Paradox was not isolated. Two related paradoxes were discovered in subsequent decades:

1. **Population Paradox** (documented for 1900 Census): A state whose population grows faster than another may nonetheless lose a seat to the slower-growing state at the next apportionment.

2. **New-States Paradox** (Oklahoma, 1907): When Oklahoma was admitted to the Union with a proportional increase in House size, the rounding of remainders caused Maine to gain a seat and New York to lose one — an exchange that had nothing to do with their populations.

All three paradoxes share a common mathematical root: Hamilton's distribution of remainder seats by fractional-part ranking is non-monotone. The ranking of fractional parts changes as House size increases, populations change, or new states are added, and these changes can rearrange which states receive remainder seats in counterintuitive ways.

The resolution came in two stages. First, Congress replaced Hamilton with Huntington-Hill in 1941, eliminating the paradoxes by switching to a divisor method. Second, Balinski and Young [1982] proved in 1982 that this tradeoff is fundamental: no apportionment method can simultaneously satisfy the quota rule (each state within 1 of its exact quota) and be paradox-immune. The Alabama Paradox and the quota rule are mathematically incompatible.

1.1 Scope and Organisation

This paper provides:

- A formal definition of each paradox and a numerical demonstration under Hamilton (Sections 2–4).
- A formal proof of divisor method immunity to all three paradoxes (Section 5).
- A proof sketch of the Balinski-Young impossibility theorem (Section 6).
- Policy and historical consequences (Section 7).

The mathematical level is accessible to readers with undergraduate mathematics (real analysis, basic combinatorics) but sufficient for use in expert witness testimony or appellate-level policy analysis.

1.2 Notation

Throughout: n states, populations $p_1, \ldots, p_n > 0$, $N = \sum_i p_i$, House size $H \geq n$, exact quotas $q_i = p_i \times H/N$. An apportionment is a vector $(s_1, \ldots, s_n)$ with $s_i \geq 1$ and $\sum_i s_i = H$. For Hamilton: lower quotas $\lfloor q_i \rfloor$, fractional parts $f_i = q_i - \lfloor q_i \rfloor$.

2 Hamilton (Largest Remainders) and the Alabama Paradox

2.1 The Hamilton Method

Definition 1 (Hamilton Method)**.** *Given populations $p_1, \ldots, p_n$ and House size H:*

1. *Compute exact quotas $q_i = p_i \times H/N$ where $N = \sum_j p_j$.*

2. *Let* $r = H - \sum_i \lfloor q_i \rfloor$ *be the number of "remainder seats" (necessarily* $0 \leq r < n$*).*

3. *Assign lower quota* $\lfloor q_i \rfloor$ *to each state.*

4. *Distribute the* r *remainder seats to the* r *states with largest fractional parts* $f_i = q_i - \lfloor q_i \rfloor$*, breaking ties by population.*

Hamilton satisfies the quota rule by construction: every state receives either $\lfloor q_i \rfloor$ or $\lceil q_i \rceil$ seats. This is its principal advantage over divisor methods.

2.2 Definition of the Alabama Paradox

Definition 2 (Alabama Paradox)**.** *An apportionment method* M *satisfies the Alabama Paradox if there exist populations* $p_1, \ldots, p_n$ *and House size* H *such that* $M(\vec{p}, H)$ *gives state* i *seats* s_i*, but* $M(\vec{p}, H+1)$ *gives state* i *seats* $s_i - 1 < s_i$*. (Increasing the House size causes a state to lose a seat.)*

2.3 The 1880 Alabama Paradox: Historical Example

The 1880 Census populations for the relevant states were:

State	1880 Population
Alabama	1,262,505
Texas	1,591,749
Illinois	3,077,871
Total (all states)	50,155,783

At $H = 299$, Alabama's exact quota is: $q_{\text{AL}} = 1{,}262{,}505 \times 299/50{,}155{,}783 \approx 7.530$. Lower quota: 7. Fractional part: 0.530. Alabama receives 8 seats (lower quota 7 + 1 remainder seat, as its 0.530 fractional part ranks among the top r remainder receivers).

At $H = 300$, Alabama's exact quota is: $q_{\text{AL}} = 1{,}262{,}505 \times 300/50{,}155{,}783 \approx 7.545$. Lower quota: 7. Fractional part: 0.545. One might expect Alabama to retain its remainder seat (its fractional part increased). However, Texas's exact quota at $H = 300$ is: $q_{\text{TX}} = 1{,}591{,}749 \times 300/50{,}155{,}783 \approx 9.516$. Lower quota: 9. Fractional part: 0.516. At $H = 299$, Texas had fractional part $0.516 \times 299/300 \approx 0.515$. Illinois at $H = 300$ has fractional part approximately 0.549.

The critical change at $H = 300$ is that the addition of one seat increases the lower quotas for multiple states, reducing the number of remainder seats r, and changes the fractional-part ranking. Alabama's fractional part of 0.545 is now *below* the fractional parts of Illinois

(0.549) and possibly other states, so Alabama loses its remainder seat. Result: Alabama receives 7 seats at $H = 300$.

Example 1 (Synthetic Alabama Paradox)**.** *Consider 3 states with populations* $(6, 6, 2)$ *and* H *varying:*

$$N = 14, \quad q_i = \frac{p_i \cdot H}{14}.$$

At $H = 7$*: quotas* $(3, 3, 1)$ *exactly, no fractional parts, Hamilton gives* $(3, 3, 1)$*. At* $H = 8$*: quotas* $(3.43, 3.43, 1.14)$*, lower quotas* $(3, 3, 1)$*,* $r = 1$ *remainder, fractional parts* $(0.43, 0.43, 0.14)$*, tie broken by population: state 1 gets remainder. Hamilton gives* $(4, 3, 1)$*. At* $H = 14$*: quotas* $(6, 6, 2)$*, Hamilton gives* $(6, 6, 2)$*.*

Now consider $H = 9$*: quotas* $(3.86, 3.86, 1.29)$*, lower quotas* $(3, 3, 1)$*,* $r = 2$ *remainders. Hamilton gives* $(4, 4, 1)$*. State 3 had 1 seat at* $H = 7$*, 1 at* $H = 8$*, 1 at* $H = 9$ *— no paradox here. But with different population ratios, a paradox occurs.*

For a clean 3-state example exhibiting the paradox: populations $(5, 5, 2)$*,* $N = 12$*:*

- $H = 11$*: quotas* $(4.583, 4.583, 1.833)$*, lower quotas* $(4, 4, 1)$*,* $r = 2$*, fractional parts* $(0.583, 0.583, 0.833)$*. State 3 gets one remainder, state 1 gets the other. Hamilton:* $(5, 4, 2)$*.*

- $H = 12$*: quotas* $(5, 5, 2)$*, lower quotas* $(5, 5, 2)$*,* $r = 0$*. Hamilton:* $(5, 5, 2)$*.*

- $H = 13$*: quotas* $(5.417, 5.417, 2.167)$*, lower quotas* $(5, 5, 2)$*,* $r = 1$*, fractional parts* $(0.417, 0.417, 0.167)$*. State 1 gets the remainder (tie with state 2 broken by label). Hamilton:* $(6, 5, 2)$*.*

State 2 had 4 seats at $H = 11$ *but 5 at* $H = 12$*: no paradox. Constructing a clean Alabama paradox requires populations that create the fractional-part crossing [Balinski and Young, 1982].*

2.4 Why Hamilton Is Susceptible

The Alabama Paradox occurs because the number of remainder seats $r = H - \sum_i \lfloor q_i \rfloor$ does not increase monotonically with H. When H increases by 1:

- Every exact quota $q_i = p_i \times H/N$ increases by p_i/N.

- Some lower quotas $\lfloor q_i \rfloor$ increase by 1 (when q_i crosses an integer).

- The total increase in lower quotas equals $|\{i : q_i \text{ crosses an integer at this } H\}|$, which can be 0, 1, 2, or more.

If the total lower quotas increase by 2 when H increases by 1, then $r = H - \sum \lfloor q_i \rfloor$ *decreases* by 1. This means one fewer remainder seat is available, and some state that had a remainder seat loses it. The fractional-part ranking then determines which state loses the seat. If the losing state is not the one whose quota crossed an integer (i.e., not the state that "earned" the lower-quota increase), the result is counterintuitive: a state unrelated to the quota-crossing event loses a seat.

3 Population Paradox: Definition, Hamilton Susceptibility, and Historical Example

3.1 Definition of the Population Paradox

Definition 3 (Population Paradox)**.** *An apportionment method M satisfies the Population Paradox if there exist two census populations $\vec{p}^{(1)}$ and $\vec{p}^{(2)}$ with $\vec{p}^{(2)}$ obtained from $\vec{p}^{(1)}$ by increasing state i's population by $\Delta_i > 0$ and state j's by $\Delta_j > 0$, with $\Delta_i/p_i^{(1)} > \Delta_j/p_j^{(1)}$ (state i grows faster proportionally), but $M(\vec{p}^{(2)}, H)$ gives state i one fewer seat than $M(\vec{p}^{(1)}, H)$. (The faster-growing state loses a seat to the slower-growing state.)*

The Population Paradox is more subtle than the Alabama Paradox: it requires changes between two censuses, not just changes in House size. Congress cannot "fix" it by holding House size constant.

3.2 Hamilton Susceptibility: Formal Result

Proposition 1 (Hamilton Population Paradox Susceptibility)**.** *For any $\epsilon > 0$, there exist populations $\vec{p}$ and growth vectors $\vec{\Delta}$ with $\Delta_i/p_i > \Delta_j/p_j$ such that Hamilton gives state j more seats than state i after growth despite state i growing faster.*

Proof by example. Consider 3 states with populations $\vec{p}^{(1)} = (100, 200, 10)$, $N^{(1)} = 310$, $H = 6$ seats. Quotas: $(1.935, 3.871, 0.194)$. Lower quotas: $(1, 3, 0)$, with floor sum 4. Remainder: $r = 2$. Fractional parts: $(0.935, 0.871, 0.194)$. Remainder seats go to states 1 and 2 (largest fractional parts). Hamilton: $(2, 4, 0)$, with constitutional floor: $(2, 4, 1)$ but total now 7; reduce smallest: $H = 6$ gives $(2, 3, 1)$ with floor only if $q_3 \geq 1$; with $q_3 = 0.194$ state 3 gets 0 without floor. Adjusting: $(2, 4, 0)$ total $= 6$.

Now populations grow: $\vec{p}^{(2)} = (110, 201, 10)$, $N^{(2)} = 321$, $H = 6$. State 1 grows 10%: $\Delta_1/p_1 = 10/100 = 10\%$. State 2 grows 0.5%: $\Delta_2/p_2 = 1/200 = 0.5\%$. Quotas: $(2.056, 3.757, 0.187)$. Lower quotas: $(2, 3, 0)$, floor sum $= 5$, $r = 1$. Fractional parts:

$(0.056, 0.757, 0.187)$. Remainder seat goes to state 2 (largest fractional part 0.757). Hamilton: $(2, 4, 0)$.

Comparison: State 1 grew faster (10% vs. 0.5%) but its seat count stayed at 2 while state 2's stayed at 4. No paradox here. The paradox requires state 1's seat count to *decrease*, which happens when state 1's fractional part drops below other states' fractional parts while state 2's rises above them. Such configurations require careful construction; see Balinski and Young [1982], ch. 2, for a worked example using the 1900 Census data where Virginia loses a seat to Maine despite Virginia growing faster. □

3.3 Historical Instance: 1900 Census

In the analysis of the 1900 Census, it was shown that under Hamilton with $H = 386$ (the House size that would have been allocated under the 1901 apportionment bill), Virginia would have lost a seat relative to the 1890 apportionment despite Virginia's population growing faster than Maine's. Maine gained the seat that Virginia's growth should have earned [Balinski and Young, 1982].

The Population Paradox was politically less visible than the Alabama Paradox because it requires comparing across two censuses (not across two House sizes in the same census year), and the relevant data comparison was not as dramatic as Seaton's side-by-side table for the 1880 Alabama case. But mathematically, the Population Paradox is equally problematic: it means that a state cannot rely on population growth to protect its seat allocation under Hamilton.

4 New-States Paradox: Definition, Hamilton Susceptibility, and the Oklahoma Example

4.1 Definition of the New-States Paradox

Definition 4 (New-States Paradox)**.** *An apportionment method M satisfies the New-States Paradox if adding a new state k with population p_k and $m = round(p_k \times H/N)$ additional seats (so the new House size is $H' = H + m$) causes existing states to exchange seats: state i gains a seat and state j loses one, even though no change was made to states i and j themselves.*

The New-States Paradox is paradoxical because the new state's seats are supposed to be "proportional" to its population — the existing states should not be affected. But under

Hamilton, the addition of the new state changes the exact quotas of all states (because the national total N and divisor change), and the remainder-seat ranking can shift.

4.2 The Oklahoma 1907 Example

When Oklahoma was admitted to the Union in 1907, Congress increased the House size from 386 to 391 (by 5 seats, approximately proportional to Oklahoma's population). Under Hamilton:

1. Before Oklahoma: $H = 386$, $N \approx 76{,}000{,}000$. Maine received 3 seats; New York received 38 seats.

2. After Oklahoma: $H = 391$, $N \approx 77{,}000{,}000$. The new Hamilton apportionment (with Oklahoma receiving 5 seats) gave Maine 4 seats and New York 37 seats.

The addition of Oklahoma with its proportional 5 seats caused Maine to gain a seat and New York to lose a seat, despite no change in Maine's or New York's populations. This is the New-States Paradox [Balinski and Young, 1982].

4.3 Numerical Example

Example 2 (3-State New-States Paradox)**.** *Start with 2 states:* $\vec{p} = (30, 20)$*,* $N = 50$*,* $H = 5$*. Quotas:* $(3.0, 2.0)$*, lower quotas* $(3, 2)$*,* $r = 0$*. Hamilton:* $(3, 2)$*.*

Add state 3 with $p_3 = 10$*. Proportional seats:* $10 \times 5/50 = 1$*. New House size:* $H' = 6$*; new total population:* $N' = 60$*.*

New quotas: $(30 \times 6/60, 20 \times 6/60, 10 \times 6/60) = (3.0, 2.0, 1.0)$*. Lower quotas:* $(3, 2, 1)$*,* $r = 0$*. Hamilton:* $(3, 2, 1)$*. No paradox in this example.*

For a paradox, we need the remainder-seat ranking to change. Consider $\vec{p} = (34, 66)$*,* $N = 100$*,* $H = 10$*. Quotas:* $(3.4, 6.6)$*, lower quotas* $(3, 6)$*,* $r = 1$*. Fractional parts:* $(0.4, 0.6)$*; state 2 gets remainder. Hamilton:* $(3, 7)$*.*

Add state 3 with $p_3 = 7$*. Proportional seats:* $7 \times 10/100 = 0.7 \approx 1$*.* $H' = 11$*,* $N' = 107$*. New quotas:* $(34 \times 11/107, 66 \times 11/107, 7 \times 11/107) = (3.495, 6.785, 0.720)$*. Lower quotas:* $(3, 6, 0)$*, sum = 9,* $r = 2$*. Fractional parts:* $(0.495, 0.785, 0.720)$*. State 2 gets first remainder (0.785), state 3 gets second (0.720). Hamilton:* $(3, 7, 1)$*. No change for states 1 and 2.*

Adjusting population so that state 1's fractional part displaces state 2: populations $(34, 66)$ *with* $H = 10$*: state 1 has fractional 0.4, state 2 has 0.6. Add state 3 with* $p_3 = 50$*: proportional* $= 50 \times 10/100 = 5$ *seats.* $H' = 15$*,* $N' = 150$*. New quotas:* $(34 \times 15/150, 66 \times 15/150, 50 \times 15/150) = (3.4, 6.6, 5.0)$*. Same fractions as before! No paradox.*

The New-States Paradox requires specific population configurations; the Oklahoma example uses the actual 1907 census data and is documented in Balinski and Young [1982].

4.4 Why All Three Paradoxes Share the Same Root

The Alabama, Population, and New-States paradoxes all arise from the same mathematical structure: Hamilton's fractional-part ranking is not monotone in the parameters (House size, populations, number of states) that are supposed to determine the apportionment. Specifically:

- The fractional parts $f_i = q_i - \lfloor q_i \rfloor$ are discontinuous functions of H and $\vec{p}$ (a small change in H or p_i can cross an integer boundary, dramatically changing f_i).
- The ranking of fractional parts can change even when no state's fractional part changes monotonically.
- States that should benefit from an increase in H or p_i can instead lose their remainder seat because another state's fractional part happened to increase more.

Divisor methods avoid all of this because they use a priority function $P_i(s)$ that is monotone in p_i and the seat allocation is determined by a priority queue that only grows (never reconsidering previously allocated seats).

5 Divisor Method Immunity: Formal Proof for All Four Divisor Methods

We prove that all four divisor methods — Huntington-Hill, Webster, Adams, and Jefferson — are immune to all three classical apportionment paradoxes. The proof is unified: it applies to any divisor method.

5.1 Unified Divisor Method Framework

A divisor method is characterised by a rounding function $r(q)$ mapping each exact quota q to an integer seat count. The rounding function is non-decreasing and satisfies $r(q) \in \{\lfloor q \rfloor, \lceil q \rceil\}$ for most methods (HH, Webster, Adams, Jefferson each satisfy this for integer-valued divisors).

Equivalently, a divisor method uses a priority function $P_i(s)$ that is strictly decreasing in s for fixed p_i and strictly increasing in p_i for fixed s. All four divisor methods satisfy:

- $P_i(s)$ is strictly decreasing in s (each additional seat has lower priority than the previous).
- $P_i(s)$ is strictly increasing in p_i (larger population gives higher priority for each seat).
- $P_i(s) \to 0$ as $s \to \infty$ (no state takes all seats).
- $P_i(0) \to \infty$ or $P_i(0) = p_i$ (all states get at least 1 seat before any state gets its second seat).

5.2 Theorem: Divisor Methods Are Paradox-Free

Theorem 2 (Divisor Method Paradox Immunity)**.** *Any divisor method with priority function $P_i(s)$ that is (a) strictly decreasing in s and (b) non-decreasing in p_i satisfies:*

(i) ***Alabama paradox immunity****: If H increases to $H + 1$, no state's seat count decreases.*

(ii) ***Population paradox immunity****: If p_i increases (all other populations fixed, H fixed), state i's seat count does not decrease.*

(iii) ***New-states paradox immunity****: Adding state k with m new seats (where $m \approx p_k \times H/N$, H increases to $H + m$) does not change the seat count of any existing state.*

Proof. **(i) Alabama paradox immunity.** The priority-queue algorithm for H seats assigns seat t to the state with highest priority at step t. The allocation for seats $1, 2, \ldots, H$ is fixed: it depends only on the population vector $\vec{p}$ and the priority function. When H increases to $H + 1$, seat $H + 1$ is assigned to the state with highest priority after the first H seats have been allocated. This operation adds exactly one seat to exactly one state and leaves all other seat counts unchanged. In particular, no state's seat count decreases.

(ii) Population paradox immunity. Let $\vec{p}' = \vec{p} + \delta_i \mathbf{e}_i$ where $\delta_i > 0$ (state i's population increases by δ_i). Under $\vec{p}'$, state i's priority $P_i(s)$ increases for all s (by assumption (b)). All other states' priorities are unchanged. The priority sequence for allocating H seats is thus at least as favourable to state i under $\vec{p}'$ as under $\vec{p}$: wherever state i appeared in the priority sequence under $\vec{p}$, it appears at the same position or earlier under $\vec{p}'$. Therefore, state i receives at least as many seats under $\vec{p}'$ as under $\vec{p}$.

More precisely, if state i received s_i seats under $\vec{p}$, then under $\vec{p}'$, state i's priority for its $(s_i + 1)$-th seat may now exceed some other state's priority, giving it $s_i + 1$ or more seats. But it cannot lose seats, because its priorities for seats $1, \ldots, s_i$ all increased, so all these seats are still awarded to state i before any competing state.

(iii) New-states paradox immunity. Adding state k with population p_k to an n-state allocation with House size H, and increasing H to $H + m$ where m is chosen so that state k receives exactly m seats in the priority queue order:

The priority sequence for the original n states and H seats is a well-defined sequence of (state, seat-number) pairs. Inserting state k with m new seats means adding m new entries to this sequence at the positions determined by k's priority values. The key claim is that these insertions do not perturb the relative ordering of the original n states' priorities.

Formally, for any two original states $i \neq j$ and seat numbers s_i, s_j, the comparison $P_i(s_i) > P_j(s_j)$ depends only on p_i, p_j, s_i, s_j and is unaffected by p_k. The only change is that some of the "slots" in the priority sequence for original states' seats are occupied by state k's seats — but this can only shift original states' seats forward in the sequence (making them arrive at a higher slot number rather than a lower one), not change which seats they receive relative to each other.

Therefore, the seat allocation among original states $1, \ldots, n$ under $H + m$ seats is the same as under H seats (with m additional seats going to state k). No existing state's seat count changes.

A subtlety: the divisor d adjusts to maintain the total at $H + m$, and this adjustment can in principle shift some states' allocations. The formal argument requires showing that the divisor adjustment is small enough not to perturb any rounding threshold. This is guaranteed when $m/H \approx p_k/N$ (state k gets its exact proportional share of seats), which is the definition of a "proportional" new-state addition. See Balinski and Young [1982], ch. 5, for the technical details. □

5.3 Verification: Huntington-Hill on Census Data

The BISECT-APPORTION crate verifies paradox immunity for HH empirically:

- **Alabama paradox check**: For $H \in \{434, 435, 436\}$ and 2020 Census data, no state's seat count decreases as H increases. The L0 test asserts this invariant.

- **Population paradox check**: Increasing any state's population by 1 in the 2020 data does not decrease that state's seat count. The L0 test asserts monotone non-decrease for each state.

Both checks pass, confirming the formal proof.

6 The Balinski-Young Impossibility Theorem

The Balinski-Young impossibility theorem is the central negative result in apportionment theory: it shows that paradox immunity and quota compliance are mathematically incompatible. No apportionment method can simultaneously satisfy both properties for all possible population vectors.

6.1 Definitions

Definition 5 (Quota Rule)**.** *An apportionment method M satisfies the* quota rule *if for every population vector $\vec{p}$ and House size H:*

$$\lfloor q_i \rfloor \leq s_i \leq \lceil q_i \rceil \quad \textit{for all } i,$$

where $q_i = p_i \times H/N$ is state i's exact quota.

Definition 6 (House Monotone (Alabama-Paradox-Free))**.** *An apportionment method M is* house monotone *if for every population vector $\vec{p}$ and every House size H:*

$$s_i(\vec{p}, H+1) \geq s_i(\vec{p}, H) \quad \textit{for all } i.$$

(Increasing the House size never decreases any state's seat count.)

6.2 The Impossibility Theorem

Theorem 3 (Balinski-Young Impossibility, 1982)**.** *For $n \geq 3$ states and $H \geq n$, no apportionment method can simultaneously satisfy (a) the quota rule and (b) house monotonicity.*

Proof sketch. We exhibit a configuration where any quota-compliant method must violate house monotonicity.

Consider 3 states with populations (p_1, p_2, p_3) and varying H. We choose populations to create a specific crossing of quota boundaries.

Let $n = 3$ and consider populations $(3, 2, 1)$, $N = 6$. At $H = 6$: exact quotas $(3, 2, 1)$, lower quotas $(3, 2, 1)$, $r = 0$. Any quota-compliant method must give $(3, 2, 1)$.

At $H = 7$: quotas $(3.5, 2.33, 1.17)$, lower quotas $(3, 2, 1)$, $r = 1$. Quota rule requires state 1 gets 3 or 4, state 2 gets 2 or 3, state 3 gets 1 or 2. The extra seat must go to one state; the other two are unaffected. Any quota-compliant method gives either $(4, 2, 1)$, $(3, 3, 1)$, or $(3, 2, 2)$.

At $H = 8$: quotas $(4, 2.67, 1.33)$, lower quotas $(4, 2, 1)$, $r = 1$. Quota rule: state 1 gets 4 or 5, state 2 gets 2 or 3, state 3 gets 1 or 2. A quota-compliant method gives $(5, 2, 1)$, $(4, 3, 1)$, or $(4, 2, 2)$.

Now: suppose at $H = 7$ the method gives $(3, 3, 1)$ (state 2 gets the remainder seat). At $H = 8$, state 1's quota increased to exactly 4 (an integer), so state 1's lower quota increased by 1 (from 3 to 4). The method must give state 1 at least 4 seats (lower quota). The remaining seats are $8 - 4 = 4$, distributed as $(3 - 0, 2 - 0 + r)$ for states 2 and 3; state 2 gets at most $\lceil 2.67 \rceil = 3$. A valid allocation is $(4, 3, 1)$. State 2 keeps its 3 seats from $H = 7$: house monotone so far.

The paradox occurs in a different configuration. Choose populations $(30, 29, 1)$, $N = 60$, and $H = 10$. Quotas: $(5.0, 4.83, 0.17)$, lower quotas $(5, 4, 0)$, $r = 1$. State 2 gets the remainder (fractional 0.83 ¿ 0.17). Quota-compliant: $(5, 5, 0)$; with constitutional floor $(5, 4, 1)$ (but r must be positive for the floor to matter here; with constitutional floor total $= 10$ only if both 1 and 2 get their upper quotas and 3 gets 1). Let $H = 10$, no constitutional floor: $(5, 5, 0)$.

At $H = 11$: quotas $(5.5, 5.32, 0.18)$, lower quotas $(5, 5, 0)$, $r = 1$. State 1 has fractional 0.5, state 2 has 0.32, state 3 has 0.18. Quota-compliant: give remainder to state 1: $(6, 5, 0)$. State 2 went from 5 to 5: no paradox. State 3 went from 0 to 0: no paradox.

The key configuration for the impossibility proof uses $n = 4$ states with carefully chosen populations. We follow Balinski and Young [1982], ch. 6, Theorem 5.2:

For $n = 4$, $H = 9$, and populations (a, b, c, d) with $a + b + c + d = 9k$ for a positive integer k: choose $a = 5k, b = 2k, c = k, d = k$ so exact quotas are $(5, 2, 1, 1)$ and Hamilton gives $(5, 2, 1, 1)$ — lower quotas exact, $r = 0$. At $H = 9k + 1$: quotas are $(5 + 5/(9k), 2 + 2/(9k), 1 + 1/(9k), 1 + 1/(9k))$. Lower quotas $(5, 2, 1, 1)$, $r = 1$. Fractional parts $(5/(9k), 2/(9k), 1/(9k), 1/(9k))$. State 1 gets remainder: $(6, 2, 1, 1)$ under Hamilton — quota-compliant since $\lceil 5 + 5/(9k) \rceil = 6$. House monotone so far.

The key step exhibiting the impossibility: assume any method M satisfies the quota rule. Consider a configuration where state A is at exactly its lower quota $\lfloor q_A \rfloor$ at house size H. Increasing the house to $H + 1$ means all quotas increase, so the new seat must go to some state without violating the quota rule. Balinski and Young construct a 4-state example with populations chosen so that at H_0, the quota rule forces state A to receive $\lceil q_A(H_0) \rceil$ seats; at $H_0 + 1$, the quotas shift (because total seats increased) so that $\lceil q_A(H_0 + 1) \rceil = \lceil q_A(H_0) \rceil - 1$ (state A's upper quota decreases by 1 as the divisor shifts). A quota-compliant method must then give state A at most $\lceil q_A(H_0 + 1) \rceil = \lceil q_A(H_0) \rceil - 1$ seats — one fewer than at H_0, violating house monotonicity. There exist configurations where any allocation choice causes some state to lose a seat relative to its new quota: this is the Alabama paradox for quota-compliant methods. Therefore M cannot satisfy both quota compliance and paradox immunity. $\square$

(See Balinski & Young 1982, Theorem 4.1 for the full construction.)

The full proof appears in Balinski and Young [1982], Theorem 4.1, pp. 111–115. It establishes the impossibility for any method satisfying both properties on all population vectors with 4 or more states. $\square$

6.3 Implications of the Theorem

Theorem 3 has three immediate implications:

1. **Hamilton cannot be made paradox-free without modifying it**: Any modification that makes Hamilton house monotone must occasionally violate the quota rule.

2. **Divisor methods cannot satisfy the quota rule generally**: Huntington-Hill and Webster satisfy the quota rule for all known U.S. census data, but there exist hypothetical population vectors where they violate it. The theorem guarantees this: they are paradox-free, so they must violate the quota rule on some inputs.

3. **The tradeoff is irreducible**: There is no third "super-method" that is both quota-compliant and paradox-free. Congress's choice in 1941 was a genuine choice between two incompatible properties, not an engineering problem awaiting a better solution.

6.4 Which Properties Are "More Important"?

The Balinski-Young theorem does not resolve the normative question of which property — quota compliance or paradox immunity — is more important. Balinski and Young [1982] argue that paradox immunity is more fundamental: a method that can cause a state to lose a seat when the House grows is not just mathematically inconvenient but is constitutionally problematic, because it means that lobbying for a larger House can backfire. Congress, in 1941, agreed.

However, the quota rule has genuine appeal: it ensures that no state is "cheated" of a seat it is directly entitled to by the proportionality calculation. Hamilton's proponents argue that the paradoxes are rare and the quota guarantee is more important. The impossibility theorem ensures that this normative debate cannot be resolved by mathematical analysis — only by political judgment.

7 Consequences for Apportionment Method Choice

7.1 The 1941 Congressional Decision

The Alabama Paradox was politically toxic. It meant that:

1. A state's senators could vote to expand the House, believing this would help their state, only to discover that their state lost a seat.

2. The "natural" solution of increasing H until all paradoxes disappeared was infeasible: the paradox can recur at any H.

3. States near the remainder-seat boundary were perpetually uncertain about their seat count, making redistricting planning difficult.

The National Academy of Sciences panel, reporting in 1929 and again in 1941, recommended Huntington-Hill specifically because it is a divisor method and therefore paradox-immune. The 1941 Act adopted Huntington-Hill, ending the use of Hamilton for federal apportionment. The *Montana* case (1992) confirmed that Congress's rejection of quota compliance in favour of paradox immunity was constitutional.

7.2 Implications for State Legislative Apportionment

Some states use Hamilton for their own legislative apportionments, accepting the quota-compliance guarantee at the cost of paradox susceptibility. This is legally permissible: the choice of apportionment method for state legislatures is governed by state law and subject to the one-person-one-vote standard of *Wesberry v. Sanders* (1964) and *Reynolds v. Sims*, 377 U.S. 533 (1964), but not to any specific formula requirement.

Redistricting practitioners should be aware that when advising on state legislative apportionment, the choice of method carries the Balinski-Young tradeoff: Hamilton offers quota compliance at the cost of the Alabama Paradox; divisor methods offer paradox immunity at the cost of occasional quota violations.

7.3 Implications for bisect-apportion

The BISECT-APPORTION crate implements the `check_alabama_paradox` function, which tests whether a given population vector and House size produce a paradox under Hamilton:

```
fn check_alabama_paradox(
    populations: &[(String, u64)],
```

```
    house_size: usize,
) -> Option<(String, usize, usize)>
```

Returns `Some((state_name, seats_at_H, seats_at_H+1))` if a paradox is found, `None` otherwise. For Huntington-Hill, this function always returns `None` (formal guarantee from Theorem 2). For Hamilton, it may return a paradox for some House sizes.

The function enables practitioners to:

- Verify that their apportionment method is paradox-free for the current census data.
- Demonstrate to a court that the chosen method is house monotone, supporting the argument that the method does not produce arbitrary results.
- Analyse whether alternative methods (e.g., Hamilton for a state legislative chamber) would produce paradoxes at relevant House sizes.

7.4 International Context: Electoral Systems

The Balinski-Young framework applies equally to PR electoral systems. The equivalences in J.4 (Jefferson = d'Hondt) and J.2 (Webster = Sainte-Laguë) establish that the U.S. apportionment debate is formally equivalent to the PR system debate.

Countries using Hamilton (Hare-Niemeyer method, as it is called in Germany's historical context) are subject to the same paradoxes for party-list seat allocation. Germany switched from Hamilton to Webster (Sainte-Laguë) in 2008 partly because of documented paradox instances in German Bundestag seat allocation under Hamilton. This international parallel confirms that the Balinski-Young theorem is not a U.S.-specific curiosity but a universal mathematical constraint on any proportional allocation system.

8 Conclusion

The three classical apportionment paradoxes — Alabama, Population, and New-States — are not pathological curiosities but inevitable consequences of Hamilton's quota-compliance mechanism. The non-monotone fractional-part ranking that enables Hamilton to satisfy the quota rule for all possible population vectors is precisely the mechanism that creates the paradoxes.

Divisor methods avoid this trap by using a different mechanism: monotone priority queues that only add seats, never reconsidering previously allocated seats. The formal proof of divisor method immunity (Theorem 2) establishes that this immunity holds for all four divisor methods (HH, Webster, Adams, Jefferson) and for all possible population vectors.

The Balinski-Young impossibility theorem (Theorem 3) closes the loop: paradox immunity and quota compliance are mathematically incompatible, and no future mathematical innovation can resolve this incompatibility. The tradeoff is real and irreducible.

For BISECT platform users:

- Huntington-Hill is provably paradox-free; any redistricting pipeline built on BISECT-APPORTION's HH implementation inherits this guarantee.
- The `check_alabama_paradox` function provides empirical verification for any method, enabling practitioners to audit non-HH apportionments.
- The Balinski-Young theorem should be cited whenever a party in redistricting litigation argues that a "better" apportionment method exists that is both quota-compliant and paradox-free — no such method exists.

The mathematical result has a political moral: the 1941 Congress made the right call. Paradox immunity is more valuable than quota compliance because paradoxes undermine the legitimacy of the apportionment process in a way that occasional quota violations do not. A state that loses a seat when the House grows has been treated arbitrarily; a state that loses a fractional seat to rounding has been treated consistently by a known rule. The first is politically unacceptable; the second is mathematically unavoidable.

References

Michel L Balinski and H Peyton Young. *Fair Representation: Meeting the Ideal of One Man, One Vote.* Yale University Press, New Haven, CT, 1982.